Peter M. Graffy, PhD, MPH

Biomedical Informaticist and Population Health Epidemiologist
Pulmonary & Critical Care Postdoctoral Fellow and Co-Founder of Straia Health
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EDUCATION

2021 – 2025 PhD, Biomedical Informatics
Northwestern University, Chicago, IL

2015 – 2017 MPH, Epidemiology, Population Health
University of Wisconsin-Madison, Madison, WI

2008 – 2012 BA, Anthropology, Women & Gender Studies, Spanish
Luther College, Decorah, IA

PROFESSIONAL POSITIONS & TRAINING

2025 – Co-Founder and Technical/Scientific Lead, Straia Health
www.straiahealth.com

2025 – T32 Postdoctoral Fellowship
Section of Pulmonary and Critical Care
Center for Computational Medicine and Clinical Artificial Intelligence
University of Chicago, Chicago, IL

2024 – 2025 American Heart Association Predoctoral Fellowship
Institute for Artificial Intelligence in Medicine
Department of Preventive Medicine
Northwestern University, Chicago, IL

2020 – 2023 Research Project Manager
Surgical Outcomes and Quality Improvement Center
Department of Surgery, Northwestern University, Chicago, IL

2015 – 2020 Research Scientist
Department of Radiology
University of Wisconsin Hospital and Clinics, Madison, WI

PUBLICATIONS

1. **Graffy PM**. The Effect of Masculinity on Sexual Health Practices among College-Age Students in the United States. Journal of the National Association of Student Anthropologists. 2011. doi:10.1002/j.sda2.20110202.0004.
2. **Graffy PM**, Loomis S, Hinshaw JL, Lee F, Kim D, Pickhardt P, Lubner M, Kitchin D, Ziemlewicz T. Pulmonary intraparenchymal blood patching decreases the rate of pneumothorax-related complications following percutaneous CT-guided needle biopsy. Journal of Vascular and Interventional Radiology. 2017. PMID: 28185770.
3. **Graffy PM**, Pickhardt PJ. Quantification of hepatic and visceral fat by CT and MR imaging: relevance to the obesity epidemic, metabolic syndrome and NAFLD. British Journal of Radiology. 2016. PMID: 26876880. **[Selected for "Best of BJR 2017"]**
4. **Graffy PM**, Lee SJ, Ziemlewicz TJ, Pickhardt PJ. Prevalence of vertebral compression fractures on routine body CT according to L1 trabecular attenuation: determining thresholds for opportunistic osteoporosis screening. American Journal of Roentgenology. 2017. PMID: 28639828.

5. Lee SJ, **Graffy PM**, Zea RD, Ziemlewicz TJ, Pickhardt PJ. Future osteoporotic fracture risk related to lumbar vertebral trabecular attenuation measured at routine body CT. Journal of Bone and Mineral Research. 2018. PMID: 29314261.
6. Pickhardt P, **Graffy PM**, Li K, Hernando D, Reeder S. Quantification of liver fat content with non-contrast MDCT: phantom and clinical correlation with MRI proton density fat fraction. American Journal of Roentgenology. 2018. PMID: 30016142.
7. Lee S, Summers R, **Graffy PM**, Pickhardt P. Population-based opportunistic osteoporosis screening: validation of a fully automated CT tool for assessing longitudinal BMD changes. British Journal of Radiology. 2018. PMID: 30433815.
8. O'Connor S, **Graffy PM**, Zea R, Pickhardt P. Does nonenhanced CT-based quantification of abdominal aortic calcification outperform the Framingham Risk Score in predicting cardiovascular events in asymptomatic adults? Radiology. 2018. PMID: 30277443.
9. Kupfer S, Lubner S, Coronel E, Pickhardt P, Tipping M, **Graffy PM**, Keenan E, Ross E, Li T, Weinberg D. Adherence to post-resection colorectal cancer surveillance at NCI-designated Comprehensive Cancer Centers. Cancer Medicine. 2018. PMID: 30338661.
10. Pickhardt P, **Graffy PM**, Said A, Jones D, Malecki K, Welsh B, Zea R, Lubner M. Multi-parametric CT for noninvasive staging of HCV-related liver fibrosis: correlation with METAVIR histopathology. American Journal of Roentgenology. 2019. PMID: 30645162.
11. **Graffy PM**, Liu J, Summers R, Pickhardt P. Deep learning-based muscle segmentation and quantification at abdominal CT: application to a longitudinal adult screening cohort for sarcopenia assessment. British Journal of Radiology. 2019. PMID: 31199670.
12. Jang S, **Graffy PM**, Lee S, Summers R, Pickhardt P. Opportunistic osteoporosis screening at routine body CT: normative L1 trabecular attenuation values from >20,000 cases. Radiology. 2019. PMID: 30912719.
13. **Graffy PM**, McKinnon S, Lee G, Remington P. Life outside: a narrative ethnographic inquiry into the determinants of homelessness. Journal of Poverty. 2018. doi:10.1080/10875549.2018.1550133.
14. **Graffy PM**, Liu J, O'Connor S, Summers R, Pickhardt P. Fully automated segmentation and quantification of aortic calcification at abdominal CT: application of a deep learning algorithm to a longitudinal screening cohort. Abdominal Radiology. 2019. PMID: 30976827.
15. **Graffy PM**, Sandfort V, Summers RM, Pickhardt PJ. Automated liver fat quantification at unenhanced abdominal CT in a longitudinal screening cohort. Radiology. 2019. PMID: 31526254.
16. Pickhardt PJ, **Graffy PM**, Zea R, Lee SJ, Liu J, Sandfort V, Summers RM. Automated CT biomarkers for opportunistic prediction of future cardiovascular events and mortality in an asymptomatic screening population. Lancet Digital Health. 2020. PMID: 32864598.
17. Pickhardt PJ, **Graffy PM**, Weigman B, Deiss-Yehiely N, Hassan C, Weiss JM. Diagnostic performance of multi-target stool DNA and CT colonography for non-invasive colorectal cancer screening. Radiology. 2020. PMID: 32779997.
18. Pickhardt PJ, **Graffy PM**, Zea R, Lee SJ, Liu J, Sandfort V, Summers RM. Opportunistic screening for metabolic syndrome in asymptomatic adults utilizing fully automated abdominal CT-based biomarkers. American Journal of Roentgenology. 2020. PMID: 32603223.
19. Pickhardt PJ, **Graffy PM**, Zea R, Lee SJ, Liu J, Sandfort V, Summers RM. Automated abdominal CT imaging biomarkers for opportunistic prediction of future major osteoporotic fractures. Radiology. 2020. PMID: 32780005.
20. Summers RM, Elton DC, Lee S, Liu J, Bagheri M, Sandfort V, Grayson PC, Mehta NN, Pinto PA, Linehan WM, Perez AA, **Graffy PM**, O'Connor SD, Pickhardt PJ. Atherosclerotic plaque burden on abdominal CT: automated assessment with deep learning on noncontrast and contrast-enhanced scans. Academic Radiology. 2020. PMID: 32958429.
21. Pickhardt PJ, **Graffy PM**, Zea R. CT-gestützte Prädiktion osteoporotischer Frakturen. Risk. 2020;297:64-72.
22. Pickhardt PJ, **Graffy PM**, Perez AA, Lubner MG, Elton DC, Summers RM. Opportunistic screening at

abdominal CT: automated body composition biomarkers for added cardiometabolic value. RadioGraphics. 2021. PMID: 33646902.

23. Pickhardt PJ, Blake G, **Graffy PM**, Sandfort V, Elton DC, Perez AA, Summers RM. Liver steatosis categorization on contrast-enhanced CT using a fully automated deep learning volumetric segmentation tool. American Journal of Roentgenology. 2020. PMID: 32936018.
24. **Graffy PM**, Summers RM, Perez AA, Sandfort V, Zea R, Pickhardt PJ. Automated assessment of longitudinal biomarker changes at abdominal CT: correlation with subsequent cardiovascular events. Abdominal Radiology. 2021. PMID: 33388896.
25. Lubner MG, **Graffy PM**, Said A, Watson R, Malecki K, Pickhardt PJ. Utility of multiparametric CT for identification of high-risk NAFLD. American Journal of Roentgenology. 2021. PMID: 33474981.
26. Sandfort V, Yan K, **Graffy PM**, Pickhardt PJ, Summers RM. Use of variational autoencoders with unsupervised learning to detect incorrect organ segmentations on CT. Radiology: Artificial Intelligence. 2021. PMID: 34350410.
27. Perez AA, Noe-Kim V, Lubner MG, **Graffy PM**, Garrett JW, Elton DC, Pickhardt PJ. Automated deep learning CT-based liver volume segmentation: defining normal and hepatomegaly for clinical practice. Radiology. 2022. PMID: 34698566.
28. Olson SL, Ingram MC, **Graffy PM**, Murphy PM, Tian Y, Samis JH, Josefson JL, Rastatter JC. Effect of surgeon volume on pediatric thyroid surgery outcomes: a systematic review. Journal of Pediatric Surgery. 2022. PMID: 34980469.
29. Deiss-Yehiely N, **Graffy PM**, Weigman B, Hassan C, Pickhardt PJ, Weiss JM. High-risk serrated lesion detection at colorectal cancer screening with multi-target stool DNA versus CT colonography. American Journal of Roentgenology. 2022. PMID: 34755523.
30. Pickhardt PJ, Perez AA, Garrett JW, **Graffy PM**, Zea R, Summers RM. Fully automated deep learning tool for sarcopenia assessment on CT: L1 versus L3 vertebral level muscle measurements. American Journal of Roentgenology. 2021. PMID: 34406056. **[Selected for "Best of MSK AJR 2022"]**
31. Guo Z, Blake GM, **Graffy PM**, Sandfort V, Summers RM, Yin L, Li W, Xu S, Chen Y, Zhou J, Shao J, Jiang Y, Qu H, Baoqing L, Cheng X, Pickhardt PJ. Hepatic steatosis: CT-based prevalence in adults in China and the United States and associations with age, sex, and BMI. American Journal of Roentgenology. 2022. PMID: 34817193.
32. Pickhardt PJ, Nguyen T, Perez AA, **Graffy PM**, Jang S, Summers RM, Garrett JW. Improved CT-based osteoporosis assessment with a fully automated deep learning tool. Radiology: Artificial Intelligence. 2022. PMID: 36204542.
33. Lee M, Zea R, Garrett J, **Graffy PM**, Summers RM, Pickhardt PJ. Abdominal CT body composition thresholds using automated AI tools for predicting 10-year adverse outcomes. Radiology. 2022. PMID: 36165792.
34. **Graffy PM**, Zimmerman L, Luo Y, Yu J, Choi Y, Zmora R, Lloyd-Jones D, Allen N. Longitudinal clustering of Life's Essential 8 health metrics in the CARDIA study. Journal of the American Medical Informatics Association. 2024. PMID: 38070172.
35. **Graffy PM**, Sunderraj A, Visa MA, Miller CH, Barrett BW, Rao S, et al. Methodological approaches for measuring the association between heat exposure and health outcomes: a global scoping review. GeoHealth. 2024. PMID: 39329101.
36. **Graffy PM**, Barrett BW, Horton DE, Kho AN, Allen NB. Critical temperature thresholds for identifying vulnerability to heat-related excess cardiovascular morbidity and mortality. Journal of the American Heart Association. 2026. PMID: 41532540.
37. **Graffy PM**, Barrett BW, Horton DE, Allen NB, Kho AN. Heat and hearts: An exposure-anchored computational phenotyping framework for assessing cardiovascular vulnerability during extreme heat. Journal of Biomedical Informatics. 2026. PMID: 41875960.
38. Garcia HC, **Graffy PM**, Barrett BW, Visa MA, Jia J, Mallen E, Mansour R, Briggs G, Ford T, Wuebbles DJ, Allen N, Horton TH, Kho A, Horton DE. Increasing urban tree canopy associated with reduced mortality: a longitudinal analysis of Chicago neighborhoods. *GeoHealth.* 2026. PMID:

42559362.

39. Raval MV, Tian Y, Schäfer W, Balbale SN, Perez MN, Ingram ME, Lehane A, Smith CJ, Sullivan GA, Reiter AJ, Hu A, Borst JM, Blake SC, Close S, Davis TL, Essner BS, Heiss KF, Huang LW, Wymore E, Paniagua-Perez D, Engelhardt KM, **Graffy PM**, Johnson JK, Lillehei CW, Gray BW, Goldstein SD, Short SS, Pandya SR, Taylor JA, Gayer C, Goldin AB, Boelig MM, Rialon KL, Jancelewicz T, Lipskar AM, Jafri M, Tracy ET, Harting MT, Sulkowski JP, Ham PB III, Vali K, Cina RA, Schindel DT, Islam S, Teeple EA, Shah SR, Gosain A, Rothstein DH, Brockel MA, Chown J, Holl JL. Implementation and effectiveness of an enhanced recovery protocol for children undergoing surgery: the ENRICH-US stepped-wedge cluster-randomized trial. JAMA Surgery. 2026. PMID: 42126839.
40. Qu S, Sillmann J, Barrett BW, **Graffy PM**, Poschlod B, Brunner L, Mansour R, von Szombathely M, Hay-Chapman F, Horton TH, Chan JL, Khedkar Rao S, Woods K, Kho AN, Horton DE. Integrating machine learning-based variable selection into Heat Vulnerability Index design. Environmental Research: Health. 2026. doi:10.1088/2752-5309/ae9633.
41. Jang H, **Graffy PM**, Barrett BW, Horton DE, Chan JL, Kho AN. Identifying heat-related diagnoses in emergency department visits among adults in Chicago: a heat-wide association study. *Science Advances.* 2026;12(35). PMID: 42647618.
42. **Graffy PM**, Chhikara K, Guleria S, Ingraham NE, Chaudhari V, Rojas JC, Buell KG, Bhavani SV, Ramesh N, Ortiz AC, Weissman GE, Barker AK, Park B, Lyons PG, Liao WT, Gao CA, Hochberg CH, Goldfarb S, Sarma A, Willmore A, Nawaz MO, Horton DE, Mutlu G, Parker WF. Association of ambient air pollution exposure and acute respiratory failure outcomes in the United States. American Journal of Respiratory and Critical Care Medicine. 2026. Under review. SSRN preprint: doi:10.2139/ssrn.6153166.
43. **Graffy PM**, Chhikara K, Guleria S, Ingraham NE, Chaudhari V, Rojas JC, Buell KG, Bhavani SV, Ramesh N, Ortiz AC, Weissman GE, Barker AK, Park B, Lyons PG, Liao WT, Gao CA, Hochberg CH, Goldfarb S, Sarma A, Willmore A, Nawaz MO, Horton DE, Mutlu G, Parker WF. Trajectory-based phenotyping of respiratory patterns in patients with lung cancer in intensive care. Critical Care Explorations. 2026. Under review.
44. **Graffy PM**, Chhikara K, Beiser DG, Del Rios M, Guleria S, Ingraham NE, Chaudhari V, Rojas JC, Buell KG, Bhavani SV, Ramesh N, Ortiz AC, Weissman GE, Barker AK, Park B, Lyons PG, Liao WT, Gao CA, Hochberg CH, Goldfarb S, Sarma A, Willmore A, Horton DE, Mutlu G, Parker WF. The critical care burden of heat-related out-of-hospital cardiac arrest: implications for climate preparedness. Journal of the American Heart Association. Under review.
45. **Graffy PM**, White M, Ahn DJ, Nawaz MO, Horton DE, Parker WF. The association of air pollution with transplant waitlist outcomes in the United States. American Journal of Transplantation. 2026. Under review.
46. Mackie J, Barrett BW, **Graffy PM**, Horton DE, Kho AN. Leveraging LLM-based weak supervision for automated detection of heat-related EMS incidents. 2026. In preparation.
47. Barrett BW, **Graffy PM**, Rasnick E, Kho A, Horton DE. Daymet and DeGAUSS: development of a method linking high-resolution gridded meteorology with geocoded healthcare encounters. GeoHealth. 2026. In preparation.

PUBLISHED CONFERENCE PROCEEDINGS/ABSTRACTS

1. Coronel E, *et al.* Adherence to colorectal cancer surveillance guidelines at three United States academic cancer centers. *Gastroenterology.* 150(4): S766-S767, 2016. [doi:10.1016/S0016-5085(16)32598-7]
2. **Graffy PM**, Ziemlewicz T, Pickhardt P. *Prevalence of vertebral compression fractures at abdominal CT according to trabecular attenuation.* Radiological Society of North America (RSNA) Annual Scientific Meeting, Chicago, IL, Nov 2016. Oral Presentation.

3. Ziemlewicz TJ, **Graffy PM**, Lee SJ, Pickhardt PJ. *Prevalence of vertebral compression fractures on routine thoracoabdominal imaging according to L1 trabecular attenuation: validation of opportunistic osteoporosis screening.* Society of Abdominal Radiology (SAR) Annual Scientific Meeting, Hollywood, FL, Mar 26-31, 2017. Oral Presentation.
4. **Graffy PM**, McKinnon S, Lee G, Remington P. *Life Outside: an investigation into the lived experiences of the Madison homeless.* University of Wisconsin Population Health Sciences Annual Poster Presentation, Madison, WI, Apr 2017. Poster Presentation.
    a. Also presented at UW-Madison Preventive Health Fellowship Meeting, Madison, WI, Feb 2018. Oral Presentation.
5. **Graffy PM**, Pickhardt P, Said A, Jones D, Malecki K, Welsh B, Zea R, Lubner M. *Multi-parametric CT for noninvasive staging of liver fibrosis from HCV: correlation with the histopathologic METAVIR score.* Presented at:
    a. Society of Abdominal Radiology (SAR) Annual Scientific Meeting, Scottsdale, AZ, Mar 4-9, 2018.
    b. American Roentgen Ray Society (ARRS) Annual Scientific Meeting, Washington, DC, Apr 21-27, 2018.
    c. Zachary Clark Radiology Research Symposium, Madison, WI, May 5, 2018. Oral Presentation.
6. **Graffy PM**, Li K, Hernando D, Reeder S, Pickhardt P. *Quantification of liver fat content with non-contrast MDCT: correlation with MRI proton density fat fraction (PDFF).* Presented at:
    a. ARRS 2018 Annual Meeting, Washington DC (Apr 21-27, 2018).
    b. Zachary Clark Radiology Research Symposium, Madison WI (May 5 2018). Oral Presentation.
7. **Graffy PM**, Lubner M, Pickhardt P. *Can multiparametric CT predict the presence of NASH and fibrosis in patients with NAFLD?* American Roentgen Ray Society 119th Annual Meeting, Honolulu HI, May 5-10, 2019. Oral Presentation.
8. Pickhardt PJ, **Graffy PM**, Zea R, Lee SJ, Liu J, Sandfort V, Summers RM. *Fully automated analysis of abdominal CT scans for opportunistic prediction of cardiometabolic events: initial results in a large asymptomatic adult cohort.* Society of Abdominal Radiology (SAR) Annual Scientific Meeting, Orlando FL, 2019. **[Roscoe E. Miller Best Scientific Paper Award.]**
9. **Graffy PM**, Jang S, Lee S, Summers M, Liu J, Pickhardt P. *Opportunistic osteoporosis screening at routine body CT: normative L1 trabecular attenuation values from >20,000 data points.* ARRS 119th Annual Meeting, Honolulu HI, May 5-10, 2019. **[Cum Laude Award for Best Oral Presentation.]**
10. **Graffy PM**, Liu J, Summers M, Pickhardt P. *Deep learning-based segmentation and quantification of muscle density and area at abdominal CT in a longitudinal adult screening cohort.* ARRS 119th Annual Meeting, Honolulu HI, May 5-10, 2019. Oral Presentation.
11. Sandfort V, Yan K, **Graffy PM**, Pickhardt PJ, Summers RM. *CT organ segmentation: use of variational autoencoders to detect incorrect segmentations in a large (>10 000 CT) dataset.* Radiological Society of North America (RSNA) Annual Meeting, Chicago IL, Nov 2019.
12. Pickhardt PJ, **Graffy PM**, Weigman B, Deiss-Yehiely N, Weiss JM. *Diagnostic performance of multi-target stool DNA test and CT colonography for non-invasive colorectal cancer screening in clinical practice.* Presented at:
    a. Society of Abdominal Radiology (SAR) Annual Meeting, Maui HI (Mar 1-6, 2020).
    b. American Roentgen Ray Society (ARRS) Annual Meeting, Chicago IL (May 3-8, 2020).
13. Pickhardt PJ, **Graffy PM**, Zea R, Summers RM. *Prediction of metabolic syndrome at abdominal CT in asymptomatic adults using automated CT-based biomarkers.* Presented at:
    • Society of Abdominal Radiology (SAR) Annual Meeting, Maui HI (Mar 1-6, 2020).
    • ARRS Annual Meeting, Chicago IL (May 3-8, 2020).
14. Deiss-Yehiely N, **Graffy PM**, Weigman B, Hassan C, Pickhardt PJ, Weiss JM. *High-risk serrated lesion detection at CRC screening: multi-target stool DNA versus CT colonography.* Digestive Disease Week 2020, Chicago IL, May 2-5, 2020.
15. **Graffy PM**, Zimmerman L, Luo Y, Yu J, Choi Y, Zmora R, Lloyd-Jones D, Allen N. *Longitudinal clustering of Life's Essential 8 health metrics: application of a novel unsupervised learning method in the*

CARDIA Study. American Medical Informatics Association (AMIA) Annual Meeting, New Orleans LA, Nov 11-14, 2023.

16. **Graffy PM**, Zimmerman L, Luo Y, Yu J, Choi Y, Zmora R, Lloyd-Jones D, Allen N. *Clustering in CARDIA: longitudinal analysis of Life's Essential 8 using a novel machine-learning method.*
    a. 17th Annual Feinberg Lewis Landsberg Research Day, Northwestern University, Chicago IL, Sep 14, 2023. Poster.
    b. 8th Annual Chicago Biomedical Informatics Data Jam, Loyola University, Maywood IL, Oct 6, 2023. Poster.
17. **Peter M. Graffy**, Ashwin Sunderraj, Maxime A. Visa, Corinne Miller, Ben Barrett, Sheetal Rao, Sara F. Camilleri, Ryan D. Harp, Chuxuan Li, Anne Brenneman, Abel Kho, Daniel E. Horton, Norrina Allen. *Climate-related heat exposure and cardiovascular health: a global scoping review.* American Heart Association EPI-Lifestyle Meeting, Chicago IL, Mar 18-21, 2024. Poster.
18. Benjamin W. Barrett, **Peter M. Graffy**, Daniel E. Horton, Abel Kho. *Daymet and DeGAUSS: development of a novel method linking high-resolution gridded meteorological data with geocoded healthcare encounters.* International Population Data Linkage Network Annual Meeting, Chicago IL, Sep 18, 2024. Rapid Oral.
19. Harrison Garcia, **Peter Graffy**, Benjamin Barrett, Maxime Visa, Ashwin Sunderraj, Jenny Jia, Daniel E. Horton. *Urban tree canopy, temperature, and mortality: a machine-learning analysis of Chicago neighborhoods.* 18th Annual Feinberg Lewis Landsberg Research Day, Northwestern University, Chicago IL, Sep 12, 2024. Poster. *Social and Public Health Section Finalist.*
20. Maxime Visa, Helen Kemprecos, Benjamin Barrett, **Peter Graffy**, Daniel Horton. *Development and validation of a novel method for healthcare data linkage to high-resolution gridded temperature and climate variables.* 18th Annual Feinberg Lewis Landsberg Research Day, Chicago IL, Sep 12, 2024. Poster.
21. **Peter Graffy**, Maxime Visa, Benjamin Barrett, Abel Kho, Daniel Horton. *Comparison of heat exposure measurement methods to determine the effect of extreme heat on cardiovascular disease mortality in Chicago.* 18th Annual Feinberg Lewis Landsberg Research Day, Chicago IL, Sep 12, 2024. Poster.
22. Mehul V. Raval, Yao Tian, Lynn Huang, Erin Wymore, Deysi Paniagua-Perez, Kirsty M. Engelhardt, Mallory N. Perez, Alison Lehane, Charesa Smith, Gwyneth Sullivan, Audra J. Reiter, Andrew Hu, Martha-Conley E. Ingram, **Peter Graffy**, *et al. Implementation yields success – results of the ENhanced Recovery In CHildren Undergoing Surgery (ENRICH-US) randomized controlled trial.* American Pediatric Surgical Association Annual Meeting, Quebec Canada, May 7-10, 2025.
23. Mehul V. Raval, Willemijn Schafer, Yao Tian, Mallory N. Perez, Lynn Huang, Alison Lehane, Erin Wymore, Deysi Paniagua-Perez, Charesa Smith, Gwyneth Sullivan, Audra J. Reiter, Andrew Hu, Martha-Conley E. Ingram, Kirsty M. Engelhardt, **Peter Graffy**, *et al. Preliminary analysis of implementation drivers in the ENRICH-US randomized controlled trial.* American Surgical Association Annual Meeting, San Diego CA, Apr 24-26, 2025.
24. **Peter Graffy**, Benjamin Barrett, Daniel Horton, Abel Kho, Norrina Allen. *Critical temperature thresholds for identifying vulnerability to heat-related excess cardiovascular morbidity and mortality in Chicago.* 19th Annual Feinberg Lewis Landsberg Research Day, Chicago IL, Sep 11, 2025. Poster.
    a. Also presented at *Illinois Weather and Public Health Response Summit*, Chicago IL, Nov 12-13, 2025. Oral Presentation.
25. **Peter Graffy**, Benjamin Barrett, Daniel Horton, Norrina Allen, Abel Kho. *Heat and Hearts: computational phenotyping of cardiovascular disease during extreme heat events using electronic health record data.*
    a. 19th Annual Feinberg Lewis Landsberg Research Day, Chicago IL, Sep 11, 2025. Poster.
    b. American Heart Association Scientific Sessions – Early Innovators Spotlight, New Orleans LA, Nov 7-10, 2025. Oral Presentation.
26. Hyojung Jang, **Peter Graffy**, Benjamin Barrett, Abel Kho. *Diagnosis-specific heat-related emergency department visits among adults in Chicago: a time-stratified case-crossover study using distributed lag nonlinear models.* 19th Annual Feinberg Lewis Landsberg Research Day, Chicago IL, Sep 11, 2025. Poster.

27. Jake Mackie, **Peter Graffy**, Benjamin Barrett. *Leveraging LLM-based weak supervision for automated detection of heat-related EMS incidents.* 19th Annual Feinberg Lewis Landsberg Research Day, Chicago IL, Sep 11, 2025. Poster.
28. **Peter Graffy**, Shan Guleria, Kaveri Chhikara, William Parker. *The REspiratory Failure Environmental Risk (REFER) Study: computational phenotyping of acute respiratory failure in ICU patients.* University of Chicago BSD Postdoctoral Symposium, Chicago IL, Sep 17, 2025. Oral Presentation. **[Best Oral Presentation Award.]**
29. Harrison Garcia, **Peter Graffy**, Benjamin Barrett, Maxime Visa, Jenny Jia, Daniel E. Horton. *Increasing tree canopy associated with reduced mortality: a longitudinal analysis of Chicago neighborhoods.* Association of Medicine and Psychiatry Annual Conference, Newport Beach CA, Oct 9-11, 2025. Poster.
30. Nawaz M.O., Goldberg D.L., Kerr G.H., Kim S-Y., Chen L., Anenberg S.C., **Graffy PM.**, Lang V., Horton D.E., Southerland V. *Modeling approaches of applying remote sensing to infer surface-level NO_2.* American Geophysical Union (AGU) Fall Meeting 2025, New Orleans LA. Oral Presentation.
31. **Graffy PM**, Chhikara K, Guleria S, Ingram NE, Chaudhari V, Rojas JC, Buell K, Ramesh N, Ortiz AC, Barker AK, Park B, Lyons PG, Liao W-T, Gao CA, Hochberg C, Horton DE, Mutlu G, Parker WF. *Cumulative air pollution exposure increases the severity of acute respiratory failure: a U.S. multicenter study.* American Thoracic Society (ATS) 2026 International Conference, Orlando FL, May 16-20, 2026. Oral Presentation. **[ATS Public Advisory Roundtable ARDS Foundation Abstract Award]** Published abstract: doi:10.1093/ajrccm/aamag162.132.
32. Bell R, **Graffy PM**, Ramdeen SL, Siringan M, Horton DE, Parker WF, Palis B, Okereke I, Odell D. *Pollution and Progression: Air Quality and Stage of Lung Cancer Diagnosis Across the U.S.* American Society of Clinical Oncology (ASCO) 2026 Annual Meeting, Chicago IL. June 1, 2026. Oral Presentation. Published abstract: doi:10.1200/JCO.2026.44.16_suppl.8003.
33. Jang H, **Graffy PM**, Barrett BW, Horton DE, Kho A. *Identifying heat-related diagnoses through linked electronic health record and environmental data: a heat-wide association study conducted in Chicago.* International Population Data Linkage Network (IPDLN) Conference 2026, Rotterdam, Netherlands. July 16, 2026. Oral Presentation. Published abstract: doi:10.23889/ijpds.v11i5.3706.
34. **Peter Graffy**, David Beiser, Kaveri Chhikara, Marina Del Rios, Shan Guleria, Nick Ingraham, Vaishvik Chaudhari, Juan Rojas, Kevin Buell, Alexander Ortiz, Sivasubramanium Bhavani, Navya Ramesh, Gary Weissman, Anna Barker, Brenna Park-Egan, Patrick Lyons, Wan-Ting Liao, Catherine Gao-Howard, Chad Hochberg, Sarah Goldfarb, Aartik Sarma, Andrew Willmore, Kathryn Sullivan, Daniel Horton, William Parker. *The critical care burden of heat-related out-of-hospital cardiac arrest: implications for climate-preparedness.* American Heart Association Resuscitation Science Symposium (ReSS) 2026, Chicago IL, Nov 7, 2026. Oral presentation. **["Best of the Best" plenary session; *abstract Or002*.]** Conference program.

AWARDS AND HONORS

A. Personal

2026		American Heart Association Early Career Investigator Award
2026		Best of the Best Plenary Session Speaker, AHA ReSS
2026		ATS Public Advisory Roundtable ARDS Foundation Abstract Scholarship
2025		American Heart Association Early Innovator Spotlight
2025		University of Chicago BSD Postdoc Travel Award
2025		University of Chicago BSD Annual Symposium Best Oral Presentation Award
2025		Chicago Wilderness Alliance George B. Rabb Force of Nature Award
2019		Society of Abdominal Radiology Roscoe E. Miller Best Paper Award
2019		American Roentgen Ray Society Cum Laude Best Oral Presentation Award
2017		Delta Omega Public Health Honor Society
2016		Best of British Journal of Radiology Award
2014	– 2018	All-American Ultimate Frisbee Association 1st Team

2012		Luther College Senior Honors Thesis Award
2008	– 2012	Presidential Scholarship
2008	– 2012	NCAA Division III All-Academic 1st Team (Men's Soccer)
2008	– 2012	Dean's List

B. Students

2023 Poster Finalist, 18th Annual Lewis Landberg Research Day, Harrison Garcia

SERVICE

A. Invited Professional and Public Lectures

- "Developing climate-adaptive healthcare for the critically ill patient," Center for Health and the Social Sciences, University of Chicago, Chicago, IL, 2026
- "Data linkage, electronic health records, and CVD," University of Hamburg, Hamburg, Germany, 2025
- "Critical temperature thresholds for heat-related cardiovascular morbidity and mortality in Chicago," University of Illinois-Chicago, Chicago, IL, 2025
- "NYUTron and LLMs in clinical decision making," Northwestern University, Chicago, IL, 2023

B. University, School, and Departmental Committees

University of Chicago
Biological Sciences Postdoctoral Association Member
The Chicago Center for Health and Environment (CACHET)
Center for Extreme Conditions and Health Excellence (CECHE)

Northwestern University
Northwestern Institute for Artificial Intelligence in Medicine: AI in Healthcare Forum Student Organizer
Health Sciences Integrated Program Student Organization Sports Chair
Defusing Disasters

C. Organization Membership

American Thoracic Society (ATS) (2025 – Present)
American Heart Association (AHA) (2023 – Present)
American Medical Informatics Association (AMIA) (2023 – Present)
American Roentgen Ray Society (ARRS) (2016-2019)

D. Grant and Fellowship Peer Review

American Heart Association (AHA), Peer Reviewer, 2027 Fellowship Bioengineering & Technology Committee (Bioengineering and Biotechnology, Machine Learning, Data Science, Artificial Intelligence, Natural Language Processing)

Deutsche Forschungsgemeinschaft (DFG; German Research Foundation), Ad Hoc External Peer Reviewer, Research Grant Proposal (2026)

E. Reviewer of Manuscripts

Osteoporosis International, Artificial Intelligence in Medicine, Journal of Computer Assisted Tomography (JCAT), Journal of Gerontology: Medical Sciences, Journal of the American Medical Informatics Association (JAMIA), Journal of the American Medical Informatics Association Open (JAMIAO), JMIR Public Health and Surveillance (JPHS), BMC Public Health, American Journal of

Critical Care Medicine (AJRCCM), Nature Communications

F. Teaching Experience

Northwestern University
2023 PUB_HLTH 303: Environmental Health Sciences (TA)
2024 PUB_HLTH 422: Intermediate Epidemiology (Guest Lecturer)

G. Research Mentoring

Name	Role / Level	Years	Department / Program	Institution
Ben Weigman	PGY3, Diagnostic Radiology	2017 – 2018	Department of Radiology	University of Wisconsin Hospital and Clinics
Nimrod Deiss-Yehiely	PGY2, Internal Medicine	2018 – 2019	Department of Radiology	University of Wisconsin Hospital and Clinics
Maxime Visa	Medical Student	2022 – 2026	Feinberg School of Medicine	Northwestern University
Harrison Garcia	Medical Student	2024 – 2026	Feinberg School of Medicine	Northwestern University
Helen Kemprecos	Medical Student	2024 – 2025	Carle School of Medicine	University of Illinois Urbana-Champaign
Jake Mackie	PhD Candidate	2024 – 2026	Health Services and Outcomes Research	Northwestern University
Mercy Berchi	PhD Candidate	2024 – 2025	Health and Biomedical Informatics	Northwestern University
Donna Hooshmand	PhD Candidate	2025	Computer Science	Northwestern University
Hyojung Jang	PhD Candidate	2025 – 2026	Biostatistics	Northwestern University
Rachel Zuo	High School Student	2025	Research Project Mentorship	—
Aarin Bothra	High School Student	2025	Research Project Mentorship	—
Raheem Bell	Thoracic Surgery Fellow	2025 - 2026	American College of Surgeons	University of Chicago

GRANTS AND OTHER FUNDING

A. Ongoing Support

Grant #: 5T32HL007605-40
PI: Mutlu & Fang
Award period: 06/30/2025 – Present
Institution/mechanism: NIH NHLBI Postdoctoral Fellowship
Title: *Research Training in Respiratory Biology*

B. Completed Support

Grant #: 24PRE1193628
Award amount: $67,388
PI: Graffy
Award period: 01/01/2024 – 6/30/2025
Institution/mechanism: American Heart Association Predoctoral Fellowship
Title: *Heat and Hearts: Computational Phenotyping Cardiovascular Disease in the Context of Extreme Heat.*

C. Under Review

Grant #: 1692273
Award amount: $200,000
PI: Graffy
Award period: 12/1/2026 – 12/1/2028
Institution/mechanism: OUCH-Int'l and Lung Cancer Research Foundation Research Grant Program
Title: *Environmental Drivers of Lung Cancer and Immunotherapy Toxicity*

Grant #: GRANT14584616
Award amount: $987,594
PI: Graffy
Award period: 12/2026 – 12/2031
Institution/mechanism: NIH/NHLBI K99-R00 Pathway to Independence Award
Title: *Environment-informed multimodal phenotyping and risk stratification of acute respiratory failure*

MEDIA

Work featured in: "Cardiac Arrest ICU Admissions Doubled on 98-Degree Days, With Older Adults and Women Showing Higher Rates." *Medical Daily*, November 5, 2026. Link here.

Work featured in: "Heatwaves Increase Risk of Cardiac Arrest, Study Finds." *HealthDay (via Powers Health)*, November 3, 2026. Link here.

Work featured in: "Cardiac Arrest ICU Admissions Rose With Outdoor Heat, Especially in Women, Older Adults." *American Heart Association Newsroom*, November 2, 2026. Link here.

Work featured in: "The Weather Tied to Higher Cardiac Arrest ICU Admissions: Study." *Becker's Cardiology*, November 2, 2026. Link here.

Work featured in: "Extreme Heat Spikes Hospital Intensive Care Cardiac Arrest Admissions." *News-Medical*, November 2, 2026. Link here.

Work featured in: "Cardiac Arrest ICU Admissions Rose With Outdoor Heat, Especially in Women, Older Adults." *Medical Xpress*, November 2, 2026. Link here.

Podcast guest: "Peter Graffy on Democratizing Climate-Health Intelligence." *Climate Proofers*, October 13, 2026. Link here.

Quoted in: "'All Heat-Related Deaths Are Preventable' but in the Chicago Area 16 Died From Heat This Summer." *Chicago Sun-Times / WBEZ*, October 3, 2026. Link here.

Work featured in: “Heat Makes Emergency Rooms Busier. Study Shows Which Illness and Injuries Rise During Hot Spells.” *Associated Press*, August 26, 2026. Link here.

Work featured in: “In Chicago ERs, Extreme Heat’s Toll Goes Far Beyond the Obvious.” *Inside Climate News*, August 26, 2026. Link here.

Work featured in: “Extreme Heat Linked to 33 Diagnoses That Go Beyond Dehydration and Heatstroke.” *Medical Xpress*, August 26, 2026. Link here.

Work featured in: “Extreme Heat Linked to 33 Clinically Reportable Diagnoses.” *HealthDay*, August 26, 2026. Link here.

Work featured in: “Study Says Protecting Mature Trees Could Save More Than 100 Lives Each Year in Chicago.” *The Cool Down*, August 23, 2026. Link here.

Work featured in: “Extreme Heat Linked to 33 Diagnoses That Go Beyond Dehydration, Heat Stroke.” *Northwestern Now*, August 23, 2026. Link here.

Work featured in: “Saving Old Trees Might Save Lives.” *The New York Times*, August 20, 2026. Link here.

Work featured in: “Protecting Mature Trees From Storms and Disease Can Reduce Chicago Deaths, Study Finds.” *Chicago Tribune*, August 17, 2026. Link here.

Work featured in: “Increasing Urban Tree Canopies Could Save Lives and Reduce Heat-Related Mortalities: Here’s How.” *The Weather (Meteored)*, August 14, 2026. Link here.

Work featured in: “Saving Old Trees May Protect People’s Health More Than Planting New Ones, Study of Chicago Neighborhoods Finds.” *Live Science*, August 13, 2026. Link here.

Work featured in: “Losing Tree Cover Could Be Costing City Residents Their Lives.” *Earth.com*, August 10, 2026. Link here.

Featured in broadcast: “More Trees, Better Health? Northwestern University Says Tree Canopy Loss Linked to Mortality.” *FOX 32 Chicago*, August 6, 2026. Link here.

Work featured in: “Newswire: How and Where to Stay Cool in Chicago.” *City Bureau*, July 15, 2026. Link here.

Work featured in: “Chicago Invests in Park Cooling Upgrades to Tackle Extreme Heat.” *Smart Cities World*, July 14, 2026. Link here.

Work featured in: “City of Chicago to Upgrade Air Conditioning at 42 Park Sites in Heat-Vulnerable Areas.” *CBS Chicago*, July 9, 2026. Link here.

Work featured in: “Mayor Brandon Johnson, Chicago Park District Bring Air Conditioning Improvements to Chicago Parks in Most Heat-Vulnerable Neighborhoods.” *City of Chicago, Office of the Mayor*, July 9, 2026. Link here.

Work featured in: “The 1995 Chicago Heat Wave Killed Hundreds. A New Tool Aims to Prevent the Next Tragedy.” *Yale Climate Connections*, September 4, 2025. Link here.

Work featured in: "West, South Side Residents Question City's Latest Extreme Heat Plan." *15 West*, July 31, 2025. Link here.

Featured in broadcast: "Northwestern University Researchers Develop Tool to Measure Communities' Vulnerability to Heat." *CBS Chicago*, July 24, 2025. Link here.

Work featured in: "30 Years After Chicago's Deadliest Heat Wave, Systemic Racism Is Still the Root Problem." *Inside Climate News*, July 17, 2025. Link here.

Work featured in: "Chicagoans Want Water Stations at Bus Stops, More Greenspace, Worker Protections Ahead of Heat Waves." *Chicago Sun-Times*, July 15, 2025. Link here.

Featured in broadcast: "Examining Effects of Extreme Heat in Chicago's Black Communities." *WTTW News*, August 26, 2023. Link here.

Featured in broadcast: "As Cities Sizzle, Mapping Projects Pinpoint the Perils of Hot Neighborhoods." *WBUR's Here & Now*, August 8, 2023. Link here.

Work featured in: "Artificial Intelligence Predicts Heart Disease Risk From CT Scans." *NIH Research Matters*, March 17, 2020. Link here.

Work featured in: "Automated CT-Based Biomarkers Can Better Predict Cardiovascular Events Than Standard Methods." *News-Medical*, March 5, 2020. Link here.

Work featured in: "Can AI Screen for Nonalcoholic Fatty Liver Disease?" *AuntMinnie*, September 19, 2019. Link here.

Work featured in: "Abdominal Aortic Calcification May Signal Future Heart Attack." *Diagnostic and Interventional Cardiology*, October 12, 2018. Link here.

Work featured in: "Routine Imaging Scans May Predict Fracture Risk in Older Adults." *ScienceDaily*, February 9, 2018. Link here.

PUBLICATION METRICS

Total Citations: **2,969**
Citations Since 2021: **2,661**
h-index: **25**
i10-index: **31**

SOFTWARE AND TOOLS

Daymet DeGAUSS Tool (2024). Open-source containerized workflow for assigning high-resolution Daymet climate data to geospatial health datasets. Available at: https://degauss.org/daymet.